ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

Self Introduction: Written Script and One-Page Profile

Harikumaran K

Aspiring Cloud Engineer | First-year CSE student aiming for a product-based company

Links	GitHub: https://github.com/harikumarank LinkedIn: https://www.linkedin.com/in/hari-kumaran-k-b74159428
Summary	I like to understand how something works before I build on top of it. When a problem blocks me, I break it into small parts and test one part at a time until it is solved. My goal is to become a professional in a product-based company. Cloud engineering looks like the best domain for me, because it needs strong basics in programming, Linux and networks, and because cloud skills will stay in demand for a long time.
Introduction	Hi, I'm Harikumaran K, a first-year B.E. Computer Science and Engineering student at St. Joseph's College of Engineering, Chennai (batch 2026–2030). I program in C and Python, use Git and GitHub, and I am now exploring Linux, computer networks and cloud computing. I speak English, Tamil and Japanese. Down the line, I want to grow into a cloud engineer, and I plan to learn Docker, Terraform, CI/CD and Kubernetes along the way.
Skills	C, Python, Git, GitHub, VS Code; learning Linux, computer networks and cloud computing
Languages	English, Tamil, Japanese
Experience	First-year student; no internship yet. Practice through C and Python coursework.
Education	B.E. CSE, 1st Year, St. Joseph's College of Engineering, Chennai (2026–2030)
Awards	None yet (first year)

Impressive Personality: Biographical Note

Walter Lee, cloud and DevOps engineer and community speaker, San Francisco

Walter Lee is a cloud and DevOps engineer based in San Francisco, California. His LinkedIn profile describes about twenty years of experience in cloud and DevOps engineering, and he has more than 42,000 followers there. He is not known for one famous product. He is known for how much he understands across modern infrastructure and for how openly he shares it with other engineers.

I chose him because cloud engineering is the domain I want to enter, and his career shows what a long, steady path in that field looks like.

Career and focus

Public profiles show that Walter works across clouds, Kubernetes, DevOps, AI and machine learning, data, and web and mobile performance. On GitHub he introduces himself as a passionate software engineer from San Francisco, with clouds and Kubernetes listed first. His speaker profile lists his talk topics as cloud native infrastructure, cloud computing on the Azure platform, and cloud security.

Cloud native infrastructure means building systems that are designed for the cloud from the start: packaged in containers, run by orchestrators such as Kubernetes, and described as code so they can be rebuilt at any time. Cloud security is the other side of the same work, since every new service also needs access control, safe configuration and monitoring.

Credentials

His speaker profile describes him as a Google Developer Expert, an AWS Community Builder and an Azure certified trainer, and it lists CKA, CKS and Terraform certifications. CKA is the Certified Kubernetes Administrator exam and CKS is the Certified Kubernetes Security Specialist exam, both run by the Cloud Native Computing Foundation as hands-on tests. The Terraform certification covers HashiCorp’s infrastructure-as-code tool. Holding recognition from Google Cloud, AWS and Azure together shows that he builds skills that transfer, instead of tying himself to one vendor.

Sharing what he learns

Walter is part of the AWS Community Builders programme and writes technical posts on the developer blogging platform dev.to. One example from June 2025 explains a change to AWS Certificate Manager that lets customers export public certificates. Posts like this take a new feature and explain it in plain language, which saves other engineers time and helps beginners follow what is changing.

Why this kind of work lasts

Almost every modern product, from a banking app to a video platform, runs on cloud infrastructure. As products grow, someone must keep them reliable, secure and affordable to run. Tools will keep changing, but the need for engineers who understand systems underneath the tools looks likely to remain. This is the kind of role I want to prepare for.

How his credentials map to my roadmap

Credential or role	What it shows	My matching step
Cloud credentials (Google Cloud, AWS, Azure)	Understanding of core cloud services and design	Foundation cloud certificate in Year 1 or 2
CKA and CKS	Hands-on Kubernetes administration and security	Learn Docker first, then Kubernetes in Year 3
Terraform certification	Infrastructure described and managed as code	Learn Terraform basics in Year 2
Community Builder and Developer Expert	Teaching and contributing to the community	Share notes on GitHub and join college clubs

What I learn from his journey

First, depth takes years. A career of about twenty years in one field shows that cloud engineering is not a passing trend, which matches my wish for a role that will last long-term.

Second, fundamentals carry over between tools. Because he works across several clouds, the common base of Linux, networking and automation matters more than any single product. That is why my first-year plan focuses on C, Python, Linux and computer networks before cloud tools.

Third, certifications and teaching work together. I plan to start with a foundation cloud certificate and aim for CKA or Terraform later, while keeping my notes and projects on GitHub so that I learn in public, as he does.

Walter Lee’s example tells me that a strong cloud career is built slowly: learn the basics well, practise by building, prove the skill, and then teach it.

Sources: Walter Lee’s public Sessionize speaker profile, LinkedIn profile and GitHub profile, and his dev.to posts as an AWS Community Builder.

Summary of Lecture Notes

NPTEL: Enhancing Soft Skills and Personality (Prof. T. Ravichandran, IIT Kanpur)

Week 1: Course Introduction and Growth Mindset

The course begins by recalling an earlier course, “Developing Soft Skills”, and then introduces mindset: the habitual attitudes through which a person reads a situation. The story of the shoe salesman who finds a place where nobody wears shoes shows how one situation can look like “no chance” to one person and a “splendid opportunity” to another.

Using Carol Dweck’s ideas, the lectures compare a fixed mindset, which avoids challenges and rejects criticism, with a growth mindset, which welcomes challenges and treats learning as lifelong. A belief such as “I am not smart enough” is a limit we have learned, not a fact. The suggested habits are to stay open like a child, tolerate critics, reject stereotypes and look for a sense of belonging.

Week 2: Time Management, Procrastination and Assertiveness

Time itself cannot be managed; what we manage are our habits and our view of time. Monochronic people prefer fixed schedules while polychronic people stay flexible. The lectures suggest logging time, finding “time leakage” such as interruptions and social media, working in one’s biological peak hours, and sorting tasks with Covey’s urgent and important quadrants and the 80/20 rule.

Procrastination comes from fear, perfectionism, laziness or feeling overwhelmed. The remedies include eating the elephant bite by bite, doing the hardest “frog” task first, keeping a clean workspace, rewarding oneself, and using the Zeigarnik effect, which makes us want to finish what we start.

Week 3: Assertiveness, Anger and Positive Thinking

Passive, assertive and aggressive behaviour are compared. Assertiveness means stating your rights without violating anyone else’s and aims at a win-win result. For anger, Aristotle’s point is that the difficult part is being angry at the right person, to the right degree and at the right time. Useful techniques include delaying the reaction, deep breathing, exercise, writing an unsent letter and turning anger into creative work.

Positive thinking is shown through the story of two hospital patients and a window that faced a blank wall. Changing negative thinking needs desire, self-examination, repetition and visualisation.

Week 4: People Skills and Being Attractive

The lectures list traits that make people dislike us, such as egotism, gossip, false promises and moodiness, and their opposites, such as humility, honesty, politeness and giving credit. Socrates’ three-filter test of truth, goodness and usefulness is offered for handling gossip. Attractiveness is described in physical, mental, emotional and spiritual layers, from neat appearance and good sleep to remembering names and helping others without expecting a return.

The lecturer stresses that these traits can be changed. A person who notices a disliked habit, such as fault-finding or breaking promises, can replace it with the opposite habit through steady practice. Being yourself, valuing yourself and valuing others are presented as the base of every good relationship at work.

Week 5: English Skills and Common Errors

English is the global common language, and avoiding common errors is part of soft skills. Using a book on common errors, the lectures practise confused word pairs, Indian English habits, subject-verb agreement, prepositions, articles and collocations, and suggest learning through exposure instead of memorising rules.

Examples include the difference between “resume” and “résumé”, between “principal” and “principle”, and the Indian English phrases “cousin brother” and “passed out”. Collective and uncountable nouns are treated carefully, because they often cause mistakes in agreement between subject and verb.

Week 6: Humour and Money

Humour reduces stress, improves health and helps communication at work; sarcasm is best avoided. Adults laugh far less than children, so ideas such as a joke board and a humour club are suggested. On money, how we treat rich and poor people, borrow and lend, and handle trust shows emotional maturity. The advice is to track income and expenses, separate need from greed, avoid impulse buying, and become valuable through excellent work.

Real wealth, the lecturer adds, also includes things money cannot buy, such as health, relationships and a good name. Examples from Michelangelo, Martin Luther King and Lincoln show how devoted work can make a person indispensable.

Week 7: Health and Love

Health affects first impressions and employability because healthy people are seen as disciplined. The lectures recommend:

- Regular exercise, preferably in the morning.
- Regular meals with a good breakfast, fruits and vegetables, enough water and fewer soft drinks.
- Seven to eight hours of sleep in a good sleep environment, away from gadgets.

Love is described as constant, free, patient and unconditional. Managing it means reading body language, letting relationships mature, handling rejection with grace and keeping romance separate from work.

Week 8: Ethics, Etiquette, Mind, Memory and Environment

Values, ethics and etiquette together make up character. Professional ethics combine written codes with society’s expectations, and business etiquette covers introductions, confidentiality, not interrupting, praising in public and reprimanding in private.

The mind can be trained through positive self-talk and clear thinking: gather facts, sleep on a decision and trust your instinct. Memory methods include association, linking numbers to images and regular mental exercise. The final lecture asks for care of the environment through saving water and electricity, avoiding plastic and planting trees, and closes with Gandhi's idea that the earth provides enough for everyone's need but not for everyone's greed.

What I will apply from this course

- **Growth mindset:** treat every error message, failed build and wrong answer in C or Python as feedback, not as proof that I am not good enough.
- **Time management:** log my study hours for a week, find where time leaks, and do the hardest task, such as a new Linux or networking topic, first.
- **Assertiveness:** say no politely to extra commitments when they threaten my weekly learning plan, and ask seniors and teachers for help clearly.
- **Communication:** keep working on common errors in English, since cloud engineers write documentation, tickets and README files every day.
- **Health and sleep:** keep regular exercise and seven to eight hours of sleep, because steady learning needs a rested mind.
- **Ethics:** respect confidentiality and honesty, which matter even more when I later work with other people's data and cloud accounts.

Taken together, the eight weeks show that technical skill alone does not make a successful engineer. Attitude, time habits, communication, health and ethics decide how far the skill can go, and I will treat them as part of my career roadmap.

SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides)

Strategic Career Roadmap: SWOC Analysis

Path to Cloud Engineer at a product-based company

Candidate: Harikumaran K | 1st Year B.E. CSE | Batch 2026 – 2030

Goal: build strong fundamentals in Linux, networks and cloud early, then grow into a long-lasting cloud engineering career

Executive Summary & Career Vision

1 Vision

Grow from a first-year CSE student into a cloud engineer at a product-based company, in a role that stays valuable for many years.

2 Objective

Learn C and Python well, then Linux, computer networks and cloud. Put fundamentals and problem-solving ahead of tool-hopping.

3 Framework

Use a SWOC matrix (Strengths, Weaknesses, Opportunities, Challenges) to draw up a 4-year milestone roadmap.

SWOC Matrix at a Glance

Strengths — Internal

C and Python basics, Git and GitHub, three languages, and a clear cloud direction

Weaknesses — Internal

Beginner in Linux, networks and cloud; no projects, certifications or internships yet

Opportunities — External

Growing cloud and DevOps demand; free student credits and learning programs

Challenges — External

Fierce competition for product-based roles; fast-changing tools; surprise cloud bills

Strengths

1	**Problem solver in training**	I enjoy breaking a problem into small steps in C and Python until it works, which is the base skill for cloud work.
2	**Programming foundation**	Comfortable with C and Python, and working with Git, GitHub and VS Code.
3	**Three languages**	English, Tamil and Japanese help me learn from varied sources and communicate with different teams.
4	**Clear direction early**	I am already exploring Linux, computer networks and cloud, with Docker, Terraform, CI/CD and Kubernetes planned.

Weaknesses

Experience gaps

I am a beginner in Linux, computer networks and cloud computing, which I am only now starting to study.

No projects on GitHub, internships or certifications so far.

Working style

Many interesting tools (Docker, Terraform, Kubernetes) can pull me into learning too much at once.

A fixed weekly plan is the habit most worth building.

Opportunities

Growing cloud demand

Cloud, DevOps and platform engineering roles keep growing in product-based companies.

Student programs

The GitHub Student Developer Pack and free student cloud credits make hands-on practice affordable.

Entry-level certifications

Foundation certifications such as AWS Cloud Practitioner or Azure Fundamentals give a structured start.

Community

Open source, hackathons, college clubs and meetups give early experience and mentors.

Challenges

Competition

Product-based companies hire selectively, so strong basics and data structures are essential.

Experience paradox

Entry roles often ask for experience that a first-year student has had little chance to build.

Tooling churn and cost

Tools change fast, and personal cloud labs risk surprise bills.

Time balance

Self-learning has to fit alongside lab records, assignments and exams.

Action Plan: Closing the Gaps

1 Knowledge gap

Use Linux every day. Study networking basics (TCP/IP, DNS, subnetting) alongside coursework.

2 Experience gap

Push every small C and Python program to GitHub with a clear README.

3 Credential gap

Prepare for a foundation cloud certification before Year 2.

Action Plan: Using Opportunities

Build in public

Keep projects on GitHub with a short architecture note in each README.

Use student programs

Run every cloud lab on student credits and set billing alarms first.

Stay cost-safe

Delete resources after each lab and check the billing page weekly.

Join the community

Attend hackathons, join a cloud or Linux club and learn from seniors.

4-Year Milestones Matrix

Year	Primary Focus	Target Deliverable	Target Credential
Year 1 (2026–27)	C, Python, Linux, networks, cloud basics	Static portfolio site hosted on the cloud	Foundation cloud certificate
Year 2 (2027–28)	Docker, Python automation, Terraform	Containerised app deployed with Terraform	Associate-level cloud certificate
Year 3 (2028–29)	Kubernetes, CI/CD	App deployed through a CI/CD pipeline	CKA or Terraform Associate
Year 4 (2029–30)	Data structures, system design, capstone	Cloud capstone project	Placement in a product-based company

Conclusion & Key Takeaways

1. **Fundamentals first**

   Strong C, Python, Linux and networking make every cloud tool easier to learn and trust.

2. **Build and show**

   Small projects on GitHub, each with a README, turn learning into proof of skill.

3. **Verdict**

   With a steady weekly plan and hands-on cloud practice, I can build a profile for a cloud engineering role in a product-based company.

Write and Improve / Speak and Improve

Speak and Improve: score sheet

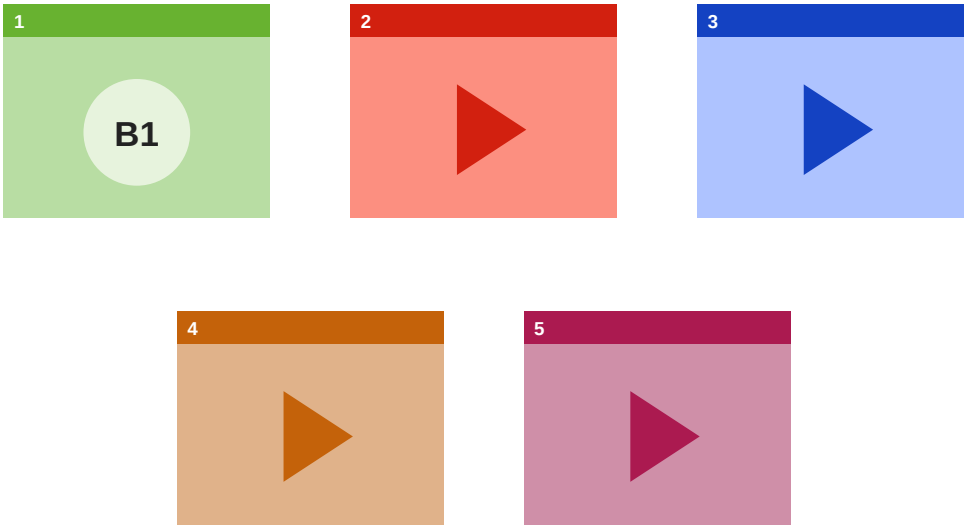
Task: Browsing on your phone

Go to this task

Check	Checked	Level
1	7 October 2026	B1

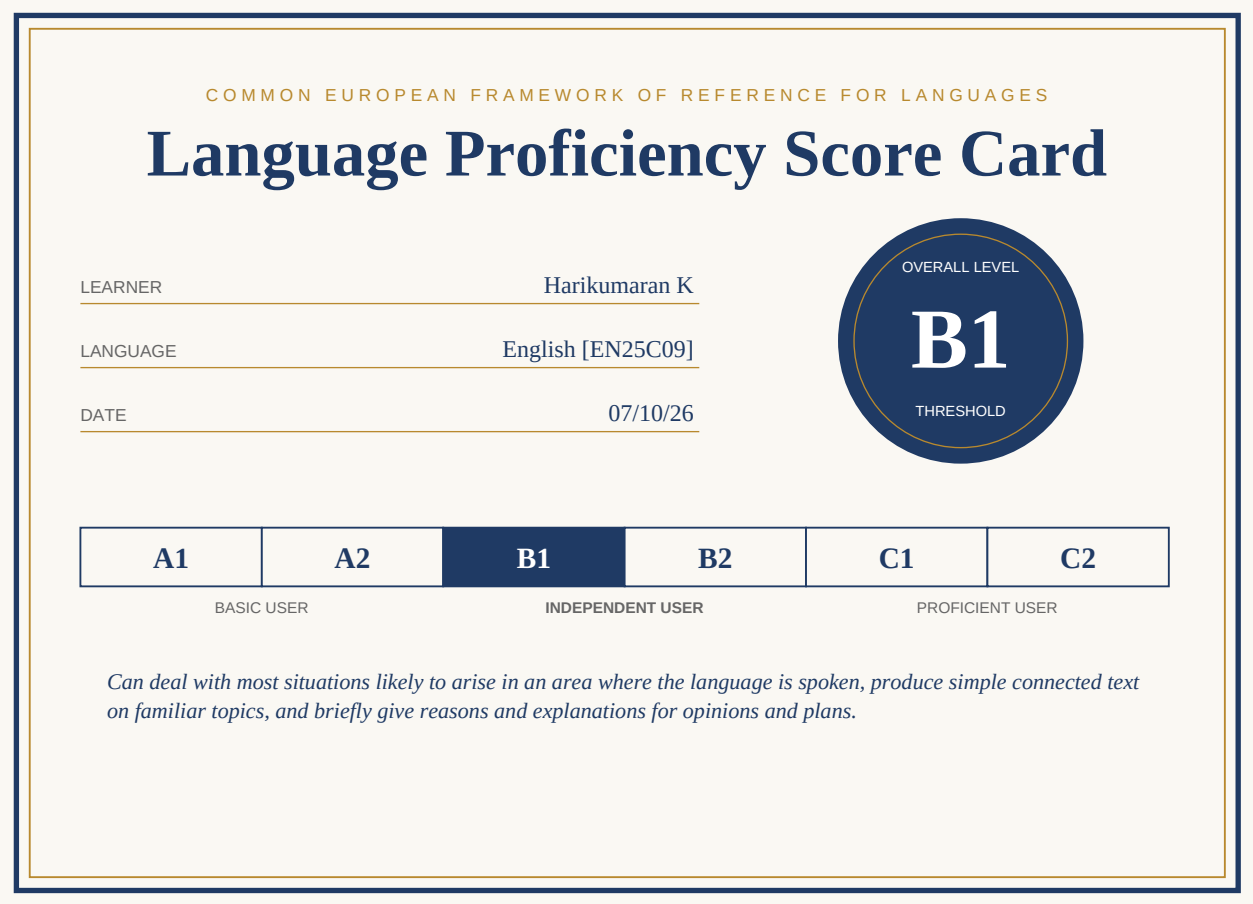
Advanced skills

These tests help you practise more advanced speaking skills. For example, summarising key points, giving explanations and discussing ideas.



Write and Improve / Speak and Improve

CEFR score card



COMMON EUROPEAN FRAMEWORK OF REFERENCE FOR LANGUAGES

Language Proficiency Score Card

LEARNER Harikumaran K

LANGUAGE English [EN25C09]

DATE 07/10/26

OVERALL LEVEL

B1

THRESHOLD

A1	A2	B1	B2	C1	C2
BASIC USER		INDEPENDENT USER		PROFICIENT USER	

Can deal with most situations likely to arise in an area where the language is spoken, produce simple connected text on familiar topics, and briefly give reasons and explanations for opinions and plans.

My CEFR level is B1 (Intermediate).

Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.

Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. monitors
2. recorded
3. stronger
4. must
5. can
6. The laboratory developed a new testing device that measures material strength with high precision.
7. rises
8. would not have failed
9. The mechanics installed dual damper springs to isolate chassis resonance.
10. optimizing
11. 50 mL of hydrochloric acid was poured into the titration flask.
12. smoothly
13. The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. The incorrect noun form is “equipments”, because “equipment” is uncountable. Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.
15. worse

Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. The engineer calibrated the pressure sensor and verified its accuracy.
2. **(Portability)** A laptop is more portable than a desktop computer because it is lighter and can be carried anywhere. **(Screen Size)** A desktop computer usually has a larger screen than a laptop, so it is more comfortable for long hours of detailed work.
3. The engineer designed a new cooling system, which reduces the temperature of industrial machines.
4. (a) If the battery voltage drops below the required level, the device will shut down automatically. (b) If the technician had checked the connections earlier, the fault would have been detected before the breakdown.
5. The solution was heated to 80°C before testing.
6. The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) The engineer received a lot of feedback on the new design. (“Feedback” is uncountable.) (b) The students received several pieces of useful advice from their teacher. (“Advice” is

uncountable.)

8. (a) The technician wears safety gloves to protect his hands. (b) The cooling fan operates continuously to prevent the motor from overheating.
9. (a) efficiently (b) carefully
10. (a) can (b) must not

Worksheet 1 – Part C (5 × 13 = 65 Marks)

Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Create a strong, different password of at least twelve characters for every account, mixing letters, numbers and symbols.
2. Turn on two-factor authentication for email, banking and social media accounts.
3. Do not open links or attachments from unknown or suspicious senders.
4. Never share OTPs, passwords, Aadhaar or bank details with anyone online.
5. Enter sensitive information only on sites whose address starts with “https” and shows a padlock.
6. Keep the operating system, browser and antivirus up to date.
7. Avoid banking and shopping on public Wi-Fi; use a trusted VPN if you must connect.
8. Report cyberbullying, online fraud and phishing at once to a trusted adult, the platform or the National Cyber Crime Helpline (1930).

Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

Therukoothu: The Street Theatre of Tamil Nadu

Therukoothu, which means “street play”, is one of the oldest folk theatre forms of Tamil Nadu. It is usually performed in villages at night, often during temple festivals, on an open stage with no fixed set. Stories are taken mostly from the Mahabharata and the Ramayana, and a performance can continue until dawn.

The actors are the main attraction. They wear bold face paint, tall crowns and bright costumes, and they sing, speak and dance while a small group plays the harmonium, mridangam and other instruments. A comic character called the kattiyakkaran introduces the play, explains the story to the audience and keeps people laughing. Villages such as Purisai in Kancheepuram district are known for keeping the tradition alive.

Today, many young people see Therukoothu as old-fashioned, and artists find it hard to earn a living from it. Modern initiatives can help. Schools and colleges can invite troupes and start drama clubs. Performances can be recorded and shared on video platforms with subtitles so that learners in cities and other countries can watch them. Cultural festivals and government support can give artists a regular stage and fair pay, and workshops can train children in the songs and make-up.

Therukoothu is a living record of our stories and values. If education, technology and public support work together, this art form will continue to be performed for the next generation.

Question 28 (b) Formal letter of request to the Principal

Harikumaran K

I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
5 November 2026

To
The Principal
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission to attend a one-day workshop on Cloud Computing and Emerging Technologies

Respected Sir/Madam,

I am a first-year B.E. Computer Science and Engineering student of your college. I have learnt that [Name of Institution] is organising a one-day workshop on Cloud Computing and Emerging Technologies on [date]. I request your kind permission to attend it.

My objective is to understand how cloud services are used in real engineering projects and to get hands-on exposure to tools that our first-year syllabus does not cover in detail. I also hope to learn from experts and meet students from other institutions.

I expect to learn the basics of cloud services, virtual machines and storage, and the latest trends in the field. After the workshop, I will share my notes with my classmates and use what I learn in my Python practice.

I request on-duty leave for the day of the workshop and permission to travel to the venue. I will make up for the classes missed and submit the participation certificate to the department. Thank you for your kind consideration.

Yours obediently,
Harikumaran K
(Student ID: 26CS1157)

Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

AI-Based Assessment versus Teacher-Based Assessment in Engineering Education

Assessment shapes how students study, so the choice between AI-based and teacher-based marking matters in engineering education. Each method has real strengths and limits.

Accuracy. An automated system applies the same rules to every code submission or objective answer and never gets tired. A teacher, however, can follow the reasoning in a long derivation or design report, which software often misses. AI is therefore reliable for objective answers, and teachers are better for open-ended ones.

Feedback time. Here AI wins clearly. Students get results and error explanations within seconds, while manual marking of a large class can take weeks. Quick feedback lets students correct a mistake while the topic is still fresh, whereas a teacher's comments arrive later but are usually more personal and motivating.

Fairness. AI does not favour anyone or mark by mood. But it can carry hidden bias from its training data, and it may reject an unusual solution that is actually valid. A teacher can notice such cases and adjust the marks.

Engineering also involves creativity, ethics and teamwork, which cannot be reduced to a score. In conclusion, neither method is perfect alone. The best approach is a blend in which AI handles routine checking quickly and teachers review complex work and guide students.

Question 30 (b) Personal email: Orientation programme

To: [friend's email address]
Subject: My first-year orientation programme at college

Dear [Friend's name],

How are you? I hope college life is going well for you too. I have been meaning to tell you about my orientation programme, so here it is.

It was held for all first-year students. The Principal gave a welcome address, and senior faculty and some alumni spoke about how to make the best of the four years. There were icebreaker games and team activities, and I made new friends on the very first day.

My favourite part was the tour of the library, computer labs, workshops and seminar halls, where staff explained how to use each facility. We were also told the rules on attendance, discipline, dress code and examinations, so I now know what is expected of us.

Overall, the programme helped me feel at home. I understood how college life works and how to plan my studies. Write back soon and tell me about your college. Take care!

Your friend,
Hari

Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize | 2. requires | 3. transmit | 4. agrees | 5. form

Exercise 2: Simple Past Tense

1. conducted | 2. deployed | 3. opened | 4. published | 5. detected

Exercise 3: Comparative Adjectives

1. higher | 2. safer | 3. faster | 4. more durable | 5. lower

Exercise 4: Modals of Obligation and Necessity

1. must | 2. must | 3. have to | 4. must | 5. has to

Exercise 5: Modals of Capability and Possibility

1. can | 2. may | 3. can | 4. can | 5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.

2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds | 2. fails | 3. engages | 4. rise | 5. registers

Exercise 8: If-Clauses (Type 3)

1. had started | 2. had lubricated | 3. would have been prevented | 4. would have tripped | 5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning | 2. replacing | 3. reducing | 4. cleaning | 5. filtering

Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely | 2. periodically | 3. rapidly | 4. firmly | 5. smoothly

Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. Incorrect: “machineries” and “equipments”. Correct: The procurement officer purchased several new pieces of machinery and equipment for the workshop.
2. Incorrect: “datas”. Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: “softwares”. Correct: We need three more software licenses to complete the finite element analysis on the model.
4. Incorrect: “informations”. Correct: The field report provided a lot of valuable information regarding the soil composition.
5. Incorrect: “apparatuses” and “glasswares”. Correct: All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse | 2. better | 3. farther | 4. less | 5. further

Worksheet 2 – Part B (Exercises 1 to 10)

Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.
4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware has to be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are also less susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they release none at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry a higher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Document databases usually provide higher horizontal query throughput than relational databases.

Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically. (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have been avoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over. (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would have been prevented.
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and an alarm will sound. (b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not have overheated.
4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately. (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have been prevented.
5. (a) If the software receives an unauthenticated API token, it will reject the request and return an error message. (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weak foundation before construction.

Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.
2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.

4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted under cryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory. (b) The team collected plenty of useful information from the field test.
2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab. (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute. (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4. (a) All electrical hardware and wiring must comply with local safety legislation. (b) The data scientist discovered several interesting insights after running the machine learning clustering.
5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises. (b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam. (b) The automated script runs every midnight to archive legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature. (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.
3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die. (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact. (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags. (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1. (a) automatically (b) flawlessly | 2. (a) accurately (b) instantly | 3. (a) continuously (b) cautiously | 4. (a) consistently (b) catastrophically | 5. (a) periodically (b) smoothly

Exercise 10: Modals

1. (a) can (b) must not | 2. (a) must (b) may | 3. (a) must not (b) could | 4. (a) should (b) must | 5. (a) can (b) must not

Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and nitrile gloves inside the laboratory at all times.
2. Open and handle strong acids and alkalis only inside the fume hood.
3. Pour acid into water slowly; never pour water into acid.
4. Do not smell, taste or touch any chemical directly.
5. Label every container clearly and never return unused chemicals to the stock bottle.
6. Wash glassware and testing equipment after every experiment to avoid contamination.
7. Keep food, drinks and open flames away from the work area.
8. Report every spill, burn or breakage to the instructor at once, and use the eyewash station or safety shower in an emergency.

Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Switch off the main supply before making or changing any connection.
2. Wear insulated gloves and safety shoes, and stand on an insulated rubber mat near live panels.
3. Do not touch exposed terminals, bare wires or rotating parts while the machine is running.
4. Ask the instructor to check your circuit before you switch on the supply.
5. Find the circuit breakers, fuses and emergency stop button before starting, and keep them accessible.
6. Remove rings, watches and metal jewellery, and avoid loose clothing, to prevent burns and short circuits.
7. Keep water and wet hands away from electrical equipment, and use only insulated tools.
8. Increase the motor load slowly, and switch off the supply and discharge the capacitors before disconnecting the circuit.

Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Wear safety glasses, safety shoes and close-fitting clothes, and tie back long hair.
2. Keep the machine door and guard closed while the spindle is running.
3. Do not touch the spindle, cutter or workpiece until the machine has stopped completely.
4. Clamp the workpiece and tool holder firmly before starting the program.
5. Remove hot chips only with a brush or chip hook and heat-resistant gloves, after stopping the machine.
6. Wipe up coolant spills at once and keep the floor dry to prevent slipping.
7. Run the program in simulation first, and keep a hand near the emergency stop button during the first cut.
8. Never leave the machine unattended or operate it without the instructor’s permission.

Worksheet 2 – Topic 2: Free Writing – Festivals and Cultural Celebrations

Question 1: A traditional harvest festival and eco-friendly practices

Pongal: A Harvest Festival That Can Stay Green

Pongal is the main harvest festival of Tamil Nadu, celebrated in mid-January to thank the Sun, the rain and the cattle for a good harvest. It lasts four days: Bhogi, Thai Pongal, Mattu Pongal and Kaanum Pongal.

On Bhogi, families clean their homes and discard old things. On Thai Pongal, they draw kolams at the doorstep and cook fresh rice and milk in a new clay pot in the open. When the milk boils over, everyone cries “Pongalo Pongal!” The pot is tied with turmeric plants and sugarcane. On Mattu Pongal, farmers decorate and worship their cattle, and Kaanum Pongal is for family visits and community games.

Eco-friendly practices can protect the spirit of the festival. Instead of burning plastic and tyres on Bhogi, communities can burn only dry leaves or compost them, which cuts smoke and pollution. Clay pots, banana leaves and areca-leaf plates can replace plastic, and kolams can be drawn with rice flour. Villages can add tree-planting drives, rainwater harvesting and cloth-bag campaigns. In this way Pongal stays joyful and traditional while being kind to nature.

Question 2: A temple car festival and sustainable measures

The Temple Car Festival: Faith, Food and a Cleaner Celebration

In many towns of Tamil Nadu, the temple car festival, or Ther Thiruvizha, is the biggest event of the year. Thousands of devotees gather to watch the deity pulled through the main streets in a huge, decorated wooden chariot called the ther.

On the car day, people of all ages hold thick ropes and pull the chariot together while drums and the nadaswaram play. People from every community take part, which makes it a strong symbol of unity. Women draw kolams on the road, families offer coconuts and flowers, and volunteers serve free food such as pongal, curd rice and payasam on leaves.

Such a large crowd also creates problems: single-use plastic litter, loud speakers and crowd safety. Organisers can ban single-use plastic and provide steel tumblers, banana leaves and paper bags. Waste-sorting points along the route and composting of food waste will keep the streets clean. Limits on loudspeaker volume reduce noise, and barricades, marked lanes, volunteers and first-aid points manage the crowd safely. With these steps the festival keeps its sacred joy and stays clean and peaceful.

Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

Folk Art and Handicraft Fairs: Heritage with a Green Future

A regional folk art and handicraft fair brings tradition and community together. Visitors watch Karagattam and Oyilattam, hear the parai drum, and see artisans make Thanjavur dolls, palm-leaf baskets, clay pots and silk sarees. The air smells of filter coffee, murukku and hot sundal.

These fairs give artisans a market and show young people the value of their heritage. Live demonstrations let children try pottery or weaving, and local food stalls support small businesses.

However, large fairs use a great deal of electricity and create a lot of waste. Solar micro-grids with rooftop panels and batteries can power lights and stalls without diesel generators. Zero-waste composting can turn food waste and leaves into manure for local farms. Organisers can use steel

plates, clay cups and cloth banners, keep separate bins for wet and dry waste, and recycle paper. When traditional skill meets clean energy, the fair protects both culture and the planet.

Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

Question 1: Technical study tour to a major hydroelectric dam

A Technical Study Tour to the Mettur Dam

Our department plans a one-day study tour to the Mettur Dam on the Cauvery River in Salem district. Forty students and two faculty members will leave by a chartered bus at 4:30 a.m., stop for breakfast on the highway, and return by night. Students will carry ID cards, water bottles and notebooks.

At the dam, an engineer will guide us through the powerhouse to see the penstocks, turbines and generators and learn how falling water produces electricity. We will also visit the control room, where operators watch water level, flow and power output, and see the spillway gates that release extra water safely.

For lunch we will try local food such as rice with sambar, fish fry and fresh mango juice. In the evening we will walk through the park beside the reservoir. The visit will show how civil, mechanical and electrical engineering work together in one project. We will wear helmets where needed and stay with our group.

Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

A Study Tour to a Solar and Wind Hybrid Energy Farm

We plan a two-day tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels supply power together. Thirty students and two faculty members will take the overnight train to Nagercoil and a hired van for the last hour, stay one night in a budget lodge and return by train the next night.

A site engineer will explain how the hybrid plant works. We will see how wind turns the blades and the generator inside the nacelle, and how the blades change angle with wind speed. In the solar section we will see photovoltaic panels, inverters and transformers feeding the grid. The control room will show how the two sources balance each other, as solar peaks in the afternoon while wind is strong in the evening and at night.

We will taste local dishes such as fish curry with rice, banana chips, avial and coconut sweets in small hotels on the way. We will wear helmets, keep away from turbine bases and follow the engineers' instructions.

Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

A Field Visit to a Seawater Reverse Osmosis Desalination Plant

Our class plans a half-day visit to a seawater reverse osmosis plant on the East Coast Road near Chennai. Fifty students and three faculty members will travel in two college buses, leaving at 8:00 a.m. and reaching the plant in about ninety minutes. We will carry ID cards and wear closed shoes.

Engineers will explain the process step by step: the intake pumps draw seawater, pre-treatment filters remove sand and particles, and high-pressure pumps push the water through membranes that

separate the salt. We will see the brine outlet, where the concentrated salt water is diluted before being released to protect marine life, and the automated control room.

Afterwards we will have lunch at a seaside restaurant with fish fry, prawn masala and rice, and filter coffee. The visit will show how engineering solves water scarcity and why careful brine disposal matters.

Worksheet 2 – Topic 4: Formal Letters of Request

Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

Harikumaran K
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
5 November 2026

To
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering, OMR, Chennai – 600 119

Subject: Request for permission and lab access after college hours to prepare for a National AI Hackathon

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon on [date] and request permission to use the computer laboratory after college hours.

Our objective is to build a simple AI-based solution to a real campus problem using Python and machine learning. We need about three weeks to design, train and test the model, and our personal laptops are not powerful enough.

We need the lab from 4:30 p.m. to 7:30 p.m. on weekdays, with Python, Jupyter Notebook, scikit-learn, Git and a stable internet connection. We promise to follow all lab rules, take care of the equipment and leave the lab clean and switched off. We will work under a faculty member's supervision if required.

Thank you for your kind consideration.
Yours obediently,
Harikumaran K (on behalf of the project team)

Question 2: Letter to the Plant Manager (industrial plant visit)

Harikumaran K
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
5 November 2026

To
The Plant Manager
[Name of Factory], Industrial Robotics and Automation Division
[Address]

Subject: Request for permission to conduct an industrial visit for 50 first-year engineering students

Respected Sir/Madam,

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai. On behalf of my class, I request permission to visit your robotics and automation factory with fifty students and two faculty members on a convenient date in [month].

We hope to learn how robots and automated systems run on a real production line, how sensors, controllers and robotic arms are combined, and how safety and quality control are managed. The visit will connect our classroom learning with practical engineering.

We will come as one group, carry ID cards, follow all safety rules and obey your staff's instructions. A short briefing and a guided tour of about two hours would be ideal. I look forward to your reply.

Yours faithfully,
Harikumaran K

Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

Harikumaran K
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
5 November 2026

To
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering, OMR, Chennai – 600 119

Subject: Request for permission and machine shop access after college hours to build an electric go-kart

Respected Sir/Madam,

Our student team has been selected for an inter-collegiate design challenge on [date], where we must build and test an energy-efficient electric go-kart. We request permission to use the machine shop after regular college hours.

We need the workshop to cut and weld the chassis, fit the motor, battery and brakes, and test the kart on the campus ground. We estimate four weeks, working from 4:30 p.m. to 7:30 p.m. on weekdays and part of the weekends.

We will work only under the supervision of the workshop instructor, wear safety glasses, gloves and shoes, follow all machine safety rules and keep the shop clean. We would be thankful for this permission.

Yours obediently,
Harikumaran K (on behalf of the team)

Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

Harikumaran K
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
5 November 2026

To
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering, OMR, Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,

I am happy to inform you that a five-member multidisciplinary team of our college has been selected to present a drone surveillance prototype at the National Defense Innovation Expo on [dates] at [venue]. We request financial sponsorship and duty leave for the event.

Our prototype uses a camera-equipped drone with image processing to monitor sensitive areas in real time. Presenting it at a national level will give us expert feedback and bring recognition to the college.

We request sponsorship for travel, accommodation, registration and prototype transport, estimated at Rs. [amount], and duty leave for [number] days. We will make up for the classes missed and submit a report after the event.

Yours obediently,
Harikumaran K (on behalf of the team)

Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

Electric Vehicles versus Conventional Internal Combustion Engine Vehicles

Electric vehicles (EVs) and internal combustion engine (ICE) vehicles answer the same need, transport, in different ways.

Battery life and manufacturing. An EV battery usually lasts eight to ten years, but replacing it is costly. An ICE vehicle has no battery problem, though its engine needs regular servicing. EVs create more emissions during production because of battery mining, yet over their lifetime they emit far less, especially when electricity comes from renewable sources.

Running cost. EVs have fewer moving parts and cheaper energy per kilometre. ICE vehicles cost less to buy and have repair shops everywhere, but rising fuel prices make them expensive to run.

Charging network. Charging stations are still limited on highways and in small towns, and charging takes longer than refuelling. An ICE vehicle can be refuelled in minutes almost anywhere.

In conclusion, EVs are cleaner and cheaper to run, while ICE vehicles are cheaper to buy and more convenient today. As batteries get cheaper and charging grows, EVs are likely to become the better choice.

Question 2: Remote/Virtual Internships versus On-Site Internships

Remote Internships versus On-Site Internships for First-Year Engineering Students

Internships link classroom theory to real work. Students can now choose between remote and on-site internships, and each has benefits and drawbacks.

Hands-on skills. On-site internships give access to machines, labs and real equipment, so students learn wiring, fabrication and testing. Remote internships mostly involve software, simulation and online tasks, which suit coding and design but cannot replace physical practice.

Flexibility. Remote internships win here, since students save travel time and cost and can balance studies. On-site work demands fixed hours and daily travel.

Mentorship and networking. On-site supervisors can guide and correct students directly, and interns meet colleagues and learn workplace culture naturally. Online mentors are often busy and reachable only through scheduled calls.

In conclusion, remote internships suit flexibility and software skills, while on-site internships suit practical learning, mentorship and networking. A first-year student can begin with a remote internship and move on-site later.

Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

Cloud Computing versus On-Premise Data Centers for Engineering Enterprises

Engineering enterprises must decide whether to run their systems on the cloud or in their own data centers. Both choices have strengths and weaknesses.

Upfront cost. Cloud services need almost no initial investment, because a company pays only for what it uses. An on-premise data center needs heavy spending on servers, cooling, power backup and buildings before any work begins.

Recurring cost. Cloud bills can grow quickly if usage is not monitored, while an on-premise setup has predictable costs once built, although it still needs staff, electricity and maintenance. The cloud suits changing workloads, and on-premise suits stable, long-term demand.

Scalability. The cloud adds computing power in minutes for simulations or product launches and reduces it afterwards. On-premise capacity is fixed, so expansion takes weeks and hardware sits idle in quiet periods.

Security and control. On-premise gives full physical control, which matters for defence or confidential designs. Cloud providers offer strong security tools and certifications, but the company depends on a third party and must follow data-location laws.

In conclusion, the cloud is better for flexibility, speed and low starting cost, and on-premise is better for control and predictable long-term expenses. Many enterprises now use a hybrid model that combines both.

Question 4: 3D Printing versus Traditional CNC Milling

3D Printing (Additive) versus CNC Milling (Subtractive) in Prototype Fabrication

3D printing builds a part layer by layer, while CNC milling cuts material away from a solid block. Each method has its own advantages.

Material wastage. 3D printing uses only the material the part needs. CNC milling removes a lot of material as chips, which raises the cost when the metal is expensive.

Speed. For single prototypes and small batches, 3D printing is faster because it needs no tooling. For larger batches, CNC milling is quicker because it can make identical parts continuously.

Finish and complexity. CNC milling gives smooth surfaces and tight tolerances. 3D printed parts show layer lines and need finishing, but a 3D printer can make hollow parts and internal channels that a cutting tool cannot reach.

In conclusion, 3D printing suits complex, low-volume prototypes with little waste, and CNC milling suits accurate, smooth, larger-batch parts. Engineers often use both.

Worksheet 2 – Topic 6: Personal Email Writing

Question 1: Email to a high school mentor/teacher about the first semester

To: [teacher's email address]
Subject: My first semester at engineering college

Dear Sir/Madam,

I hope you are well. Your guidance in school gave me the confidence to begin engineering, and I would like to tell you how my first semester is going.

The biggest change is independent campus life. Nobody reminds us of deadlines now, so I am learning to plan my own study hours and take responsibility for my work. It was hard at first, but I am getting used to it.

In the labs we wrote our first programs in C and Python. Seeing my own code run correctly gave me great satisfaction and showed me how different practical learning is from reading theory.

For the end-semester exams, I have made a timetable that gives time to every subject, revise notes every weekend and solve previous question papers. I would be happy to receive any advice from you for the coming semesters.

With respect and regards,
Harikumaran K

Question 2: Email to a high school science teacher about the first term

To: [science teacher's email address]
Subject: Reflections on my first term in engineering

Dear Sir/Madam,

I hope you are keeping well. A few months have passed since I joined engineering college, and I wanted to share my first-term experience with you, since you made science exciting for me.

The lab coursework is very different from school. We now handle components, tools and software ourselves. I made mistakes at the beginning, but each one taught me something, and I am becoming more careful and confident.

I also joined a technical club activity, where I learned about teamwork and debugging from my seniors. For the coming exams, I will finish revising each unit two weeks early, practise numerical problems daily and repeat the lab experiments so that I can answer viva questions without fear.

Yours sincerely,
Harikumaran K

Question 3: Email to an elder sibling/alumnus seeking advice

To: [sibling's or alumnus's email address]
Subject: Need your advice on balancing studies and technical clubs

Dear [Name],

How are you? I hope college and work are going well. I am writing because I need your advice on something I find difficult.

My day starts at 6:30 a.m., and I attend classes until 4:30 p.m. After that I go to my technical club for about two hours, and then I try to study and finish assignments. By the time I am free, I am very tired. My biggest challenge is the weekly lab record, which piles up and is often finished at the last minute.

Since you have faced the same situation, could you tell me how you balanced academics and clubs? I would also like your tips for semester exam revision, such as how to plan a timetable, which subjects to start with and how to use previous question papers.

Thank you in advance, and take care.
Your younger brother,
Hari

Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

Question 1: 4-day tour to an aerospace and satellite launch station

A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members, travelling by chartered train coach and local bus to keep the cost reasonable.

Budget (per student): train fare Rs. 1,300; local bus Rs. 350; accommodation for three nights Rs. 1,500; food Rs. 1,200; entry and miscellaneous Rs. 400. The total is Rs. 4,750 per student, or Rs. 1,90,000 for the group of forty.

Day 1: overnight travel, arrival, check-in and orientation briefing. **Day 2:** launch complex viewing gallery, rocket assembly area and mission control centre. **Day 3:** satellite integration and testing laboratory and the space museum, followed by an interaction with engineers on careers in aerospace. **Day 4:** group discussion, tour report preparation and the return journey.

Safety: carry ID cards and keep them visible, follow the security staff, avoid photography in restricted areas, wear closed shoes, drink plenty of water and stay with the group.

Question 2: 4-day tour to a major commercial port and maritime dry-dock facility

A 4-Day Industrial Exposure Tour to a Commercial Port and Dry-Dock Facility

Forty first-year students and two faculty members will visit a major commercial port and a dry-dock facility over four days, travelling in a hired bus and staying in a budget hotel near the port.

Budget (per student): travel Rs. 1,500; stay for three nights Rs. 1,800; food Rs. 1,200; entry passes and miscellaneous Rs. 400. The total is Rs. 4,900 per student, or Rs. 1,96,000 for the group.

Day 1: travel, check-in and short orientation. **Day 2:** briefing at the port authority and a visit to the container terminal to see cranes, conveyors and tracking systems. **Day 3:** the dry-dock, where ships are lifted and repaired, with an engineer's talk on ship stability. **Day 4:** control tower and vessel traffic system, feedback session and return.

Dockyard safety: wear helmets, safety shoes and high-visibility jackets; stay away from moving cranes, cargo and the dock edge; follow the guides, never touch machinery and keep emergency contacts and first-aid supplies ready.

Question 3: 4-day tour to an electric locomotive manufacturing works and railway signalling center

A 4-Day Study Tour to an Electric Locomotive Works and a Railway Signalling Center

Thirty-five students and two faculty members will visit an electric locomotive works and an automated railway signalling center. We will book sleeper-class train seats for both journeys and use a local bus for the visits.

Budget (per student): rail tickets Rs. 1,400; local transport Rs. 300; accommodation for three nights Rs. 1,500; food Rs. 1,200; entry and miscellaneous Rs. 300. The total is Rs. 4,700 per student, or Rs. 1,64,500 for the group.

Day 1: overnight train and arrival. **Day 2:** the locomotive works, to see shell fabrication, wiring, traction motor assembly and final testing. **Day 3:** the signalling center, where engineers explain how sensors, track circuits and computers control train movement. **Day 4:** interaction with engineers, tour report and return.

Site safety: wear helmets, safety shoes and closed clothing on the shop floor; never walk on the tracks; keep a safe distance from cranes and live electrical equipment; do not touch machines or signal equipment; and report any problem to the faculty coordinator.

Project Proposal: Ideation

CloudGuard: Student Cloud Cost and Resource Watcher, Technical Specification

Harikumaran K | CSE (Batch 2026–2030) | Draft v1, November 5, 2026

Status: draft. Items marked [default] are working assumptions that can change. Items marked [open] still need a decision.

1. Overview

CloudGuard is a small command-line tool with a generated status page that helps students run cloud labs without surprise bills. It reads account information with read-only access, finds resources that were left running, compares spending with a monthly budget and warns the student before the budget is crossed.

The aim is to let a beginner practise cloud skills safely, and to let me practise Python, Linux, Git, cloud services and later Docker, Terraform and CI/CD on one real, useful project. The tool is written in Python. The first supported provider is AWS [default], and the provider layer is kept separate so that Azure can be added later [open]. Any use of AI help will be stated in the README, and I must be able to explain every file.

2. Goals and non-goals

2.1 Goals

- Show clearly which cloud resources are running and what they cost per day.
- Warn when forecast spending will cross the monthly budget, before it happens.
- List forgotten resources such as idle virtual machines, unattached disks and unused public IP addresses, with a clear reason for each.
- Run on Linux, macOS and Windows with a few commands and no installer.
- Keep the core rules testable without a cloud account.
- Be fully documented so a teacher or a new teammate can run, read and extend it.

2.2 Non-goals

- It never creates, changes or deletes cloud resources. It only reads and reports.
- It does not replace the provider’s own billing alarms, which are still the first safety step.
- No user accounts, online service or database server in the first version.
- No machine learning. Forecasts use simple averages that a student can explain.
- No packaged executables. Users run the tool from source.

3. Target users and design feel

- **Users:** first- and second-year students running cloud labs on free tiers or student credits, with little or no cloud experience.
- **Feel [default]:** calm, plain and educational. Every warning explains why it appeared and what to do next, so the user thinks “I see what I forgot”, not “the tool is nagging me”.

- **Inspiration:** the billing dashboards of cloud providers and simple Linux command-line tools. No code or design is copied.

4. Technology stack

Item	Decision
Language	Python 3 (minimum tested version recorded in the README)
Cloud library	boto3 for AWS, version pinned in requirements.txt [default]
Configuration	config.json validated on load
Testing	pytest (development only, in requirements-dev.txt)
Report output	Terminal text and one generated static HTML page
Version control	Git and GitHub, commit after every working step
Platforms	Linux first, then macOS and Windows
Later phases	Docker image, GitHub Actions pipeline and Terraform for hosting the status page

4.1 How users run the tool

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
python3 cloudguard.py scan --config config.json
```

On Windows the activation command is `.venv\Scripts\activate`. The README lists all three variants.

5. Repository structure

```
cloudguard/
  cloudguard.py          entry point
  requirements.txt       boto3 only, pinned
  requirements-dev.txt   pytest
  README.md              full documentation
  LICENSE                chosen later [open]
  config.example.json    sample settings, no secrets
  core/                  rules and models, no cloud imports
  providers/             aws.py and fake.py (test data)
  report/                terminal.py and html.py
  tests/                 pytest tests for core and report
  samples/               fake account snapshots as JSON
  output/                generated reports, gitignored
```

Rule of thumb: **providers** fetch facts, **core** decides what they mean, **report** shows the result, and **samples** let anyone test without an account.

6. Architecture

6.1 Layers

Layer	Responsibility	Rules
providers/	Calls the cloud API and returns a plain snapshot: resources, tags, daily costs	Read-only calls only. Never logs credentials
core/	Models, rules engine, budget forecast	Never imports boto3. Fully unit-tested
report/	Terminal table and HTML status page	Contains no rules, only display
config	Budget, thresholds, region, enabled checks	Validated on load, defaults if invalid

6.2 Main flow

1. Load and validate `config.json`.
2. Ask the provider for a snapshot of resources and recent daily costs.
3. Run every enabled check on the snapshot and collect findings.
4. Forecast month-end spending from the average of recent days.
5. Print the report and, if asked, write the HTML status page to `output/`.

6.3 Error handling

Missing credentials, a missing config file, a network failure or an API limit must never crash the tool with a confusing trace. It prints one clear message, explains the fix and exits with a non-zero code. If one check fails, the other checks still run and the report says which one was skipped.

7. Commands

Command	What it does
scan	Collects a snapshot, runs all checks and prints the findings
report --html	Same as scan, and also writes a static HTML status page
budget	Shows spending so far, the daily average, the forecast and days left
explain <check>	Explains a check in plain language with a cloud-safety tip
demo	Runs everything on sample data, so a beginner can try the tool with no account

7.1 Sample terminal output (demo data)

```
CloudGuard scan  |  region ap-south-1  |  budget 10.00 USD
-----------------------------------------------------------
Spent so far:   4.20 USD   (day 12 of 30)
Daily average:  0.35 USD
Forecast:       10.50 USD  -> CRITICAL (105% of budget)

[WARN] idle_vm  i-demo-01  CPU 2% for 4 days
       Why: the machine is running and billed but barely used.
       Do:  stop it if the lab is finished.
[WARN] unattached_volume  vol-demo-07  8 GB, 6 days old
       Why: the disk is not attached to any machine but still costs money.
       Do:  delete it after checking that you no longer need it.

2 findings, 1 budget alert.
```

The tool only suggests actions. The student decides and makes any change in the provider's own console.

8. Checks (rules)

Each check has an id, a name, a severity, a threshold in the config, a plain-language reason and a suggested action.

Check	Finds	Default threshold
idle_vm	Virtual machine that is running but has very low CPU use	CPU under 5% for 3 days
unattached_volume	Disk not attached to any machine, still billed	Any, older than 1 day
unused_public_ip	Public IP address not linked to a resource	Any
old_snapshot	Backup snapshots that are no longer needed	Older than 30 days
budget_forecast	Forecast spending will cross the monthly budget	Forecast over 80% of budget = warning, over 100% = critical
missing_tag	Resource without an owner or lab tag	Tag "lab" absent

8.1 How each check works

- **idle_vm:** reads the daily average CPU use from the provider's monitoring service for every running machine. If it stays under the threshold for the set number of days, the machine is reported.
- **unattached_volume:** lists all disks and reports those with no attachment. The age is shown so a student can tell a new disk from a forgotten one.
- **unused_public_ip:** lists public addresses and reports those that are not linked to a running resource, because some providers bill for idle addresses.
- **old_snapshot:** lists backups older than the limit and shows their size, since stored backups keep costing money.
- **budget_forecast:** averages the last seven days of cost, multiplies by the days left in the month, and adds the amount already spent. This simple method can be explained and checked by hand.

- **missing_tag:** reports resources without a “lab” tag, because untagged resources are the easiest to forget.

9. Data and configuration

All limits live in `config.json`, so they can be tuned without changing code.

```
{
  "provider": "aws",
  "region": "ap-south-1",
  "monthly_budget": 10.0,
  "currency": "USD",
  "checks": { "idle_vm": true, "unattached_volume": true, "budget_forecast": true
},
  "thresholds": { "idle_cpu_percent": 5, "idle_days": 3, "snapshot_age_days": 30 }
}
```

Credentials are never stored in this file. They come from the provider’s standard environment variables or credential profile.

9.1 Data model

Record	Fields
Resource	id, type, name, region, state, created date, tags, estimated daily cost
DailyCost	date, service, amount
Snapshot	taken-at time, list of resources, list of daily costs, notes about skipped data
Finding	check id, resource id, severity, plain-language reason, suggested action

These records are plain Python data classes. The fake provider returns the same records from JSON files in `samples/`, which is what makes the core testable without a cloud account.

10. Security rules

- Use a dedicated read-only user or role with only the permissions the checks need.
- Never commit keys. The repository has a .gitignore for output files and local config, and a check for secrets before each push.
- Never print credentials or full account identifiers in logs or reports.
- Turn on the provider’s own billing alarm first. CloudGuard is a second line of defence.

11. Testing and quality

- Every rule in `core/` has a pytest test using the fake provider and sample snapshots, so tests need no cloud account.
- Tests must pass before a phase is called finished.
- A manual checklist for each phase: missing config, wrong region, no credentials, empty account and a large account.
- A “demo” run on sample data must always work on a clean machine.

10.1 Permissions the tool needs

- List and describe virtual machines, disks, public addresses and snapshots.

- Read monitoring metrics such as CPU use.
- Read daily cost data.
- Nothing that creates, modifies or deletes anything. The README gives an example read-only policy and explains each line.

11.1 Planned test cases

Case	Input	Expected result
Empty account	Snapshot with no resources	Report says “no findings” and still shows spending
Idle machine	CPU 2% for 4 days, threshold 3 days	One idle_vm finding with a reason
Busy machine	CPU 40% every day	No finding
Forecast over budget	Average 0.35 per day, budget 10, day 12	Critical budget alert (forecast 10.50)
Bad config	Negative budget or unknown check name	Clear error message and safe defaults
No credentials	Provider cannot log in	One-line fix message, no stack trace

12. Documentation requirements

The README is updated after every phase. It contains a description and screenshot, a 30-second quick start, commands and settings, an explanation of each check, the architecture, a plain-language tour of every folder, how to add a new check or provider, how to run tests, the security notes, the disclosure of any AI help, known issues, roadmap, changelog and license. Code uses type hints and docstrings, and comments explain why, not what.

13. Use of AI and disclosure

- If an AI assistant helps with code or explanations, the README and the submission will say where it was used.
- Every file must be read and understood by me before it is committed, and I must be able to explain it.
- Learning comes first: I write the core rules myself and use help only for review and for understanding unfamiliar library calls.

14. Development plan

Phase	Content	Target
1	Repository, config loading, fake provider, sample data, “demo” command	Week 1–2
2	Core rules and budget forecast with pytest tests, terminal report	Week 3–4
3	Read-only AWS provider, tested on a small free-tier account with a billing alarm	Week 5–6
4	HTML status page, explain command, README v1	Week 7
Freeze	No new features. Clean-machine test, screenshots, final README	Week 8
5 (stretch)	Docker image and a GitHub Actions pipeline that runs tests	After freeze
6 (stretch)	Host the status page with Terraform and a scheduled daily scan	After freeze

If time runs short, phases 5 and 6 are cut first. The full tool works without them.

15. Risks and mitigations

Risk	Mitigation
Scope grows too large for a first-year schedule	Phased plan, strict freeze and stretch items cut first
Cloud API calls cost money or hit limits	Cache results, run on a small account and read provider pricing notes before using billing APIs
A bug exposes credentials	Read-only role, no keys in files, secret scan before push
Forecast is inaccurate early in the month	Show a clear “low confidence” note until enough days of data exist
I do not yet understand a cloud service well	Study the service and write a short note before coding its check
Free-tier rules change	Keep thresholds in config and link to the provider’s current pages in the README

16. Open questions

1. Which provider should come first: AWS or Azure? [open]
2. Is a student-credit account enough for realistic testing? [open]
3. Should alerts be sent by email in a later version?
4. Which license will the repository use?
5. Should the status page be public or only visible to the owner?

17. Glossary

- **Free tier:** a limited amount of cloud usage a provider offers at no cost.
- **Snapshot (in this tool):** a plain record of resources and costs at one moment.
- **Read-only access:** permission to view resources without changing them.
- **Idle resource:** a resource that is running and billed but barely used.

- **Infrastructure as Code (IaC):** describing cloud resources in files so they can be created again exactly.
- **CI/CD:** automatic testing and delivery of code after each change.

18. Acceptance criteria

1. “demo” runs on a clean machine with only the two documented install steps.
2. All core tests pass and the fake provider covers every check.
3. A scan of a real free-tier account lists a deliberately forgotten machine and disk.
4. The forecast alert appears before a test budget is crossed.
5. The tool never changes anything in the cloud account, which is confirmed by reviewing the read-only policy.
6. The README lets a classmate install and run the tool without my help.

19. What I will learn in each phase

Phase	Skills practised
1	Python packages and virtual environments, JSON handling, Git branches and commits, Linux terminal use
2	Writing testable functions, pytest, date and time handling, simple statistics
3	Cloud basics: regions, virtual machines, disks, public addresses, monitoring and billing, and read-only access policies
4	Generating HTML, writing clear documentation and screenshots
5	Docker images and a GitHub Actions workflow
6	Terraform basics and scheduled jobs

20. Future ideas (after version 1)

- Support a second provider, such as Azure, through the same provider interface.
- Send a daily summary by email or a messaging app.
- Add a simple chart of daily cost to the HTML status page.
- Let a teacher view a combined, anonymous summary for a whole lab group.
- Package the tool as a Docker image so that it runs the same way on every machine.

21. Reading list

- The provider’s current free-tier and billing documentation, to confirm limits before each lab.
- The cost optimization ideas in the AWS Well-Architected Framework, to understand why idle resources matter.
- The FinOps Foundation’s introductory material on managing cloud spending.
- The official Python and pytest documentation for packaging and testing.
- The Twelve-Factor App guidelines, especially the advice on keeping configuration and secrets out of code.

Every item above will be re-checked against the latest official pages before it goes into the README, because cloud pricing and free-tier rules change.

22. Weekly learning log (template for the 8-week plan)

Week	Planned work	Evidence to keep
1	Create repository, virtual environment, config loader	First commits and a short README
2	Fake provider, sample JSON data, demo command	Demo output screenshot
3	Models and idle, disk and IP checks with tests	Passing pytest run
4	Budget forecast and terminal report	Test cases for forecast
5	Read-only provider and permissions policy	Policy file and notes on each permission
6	Real scan on a small free-tier account with a billing alarm	Redacted scan output
7	HTML status page and explain command	Page screenshot
8	Freeze, clean-machine test, final README	Tested install steps

LinkedIn and GitHub Profiles

Part A: LinkedIn profile

Harikumaran K

Aspiring Cloud Engineer | CSE Student (Batch 2026–2030) | C, Python & Linux | github.com/harikumarank

About

I'm a first-year Computer Science Engineering student who likes to understand how things work before building on top of them. I work in C and Python, use Git and GitHub, and I am now learning Linux, computer networks and cloud computing.

My goal is a long-lasting career as a cloud engineer in a product-based company. I plan to learn Docker, Terraform, CI/CD and Kubernetes step by step, and to keep my projects and notes on GitHub. I speak English, Tamil and Japanese.

Focus areas: C • Python • Linux • Computer networks • Cloud fundamentals • Git and GitHub

Featured Projects

CloudGuard (planned)

A command-line tool and status page that finds forgotten cloud resources and warns students before their monthly budget is crossed. In design; see Experiment II-7.
Tech: Python, Linux, Git, cloud APIs | github.com/harikumarank

More repositories will be added as I finish my C and Python practice programs.

Education

Computer Science & Engineering (B.E.), St. Joseph's College of Engineering, Chennai. 1st Year, Batch 2026 – 2030.

Honors & Awards

None yet (first year).

Skills

Area	Skills
Programming languages	C, Python
Spoken languages	English, Tamil, Japanese
Tools	Git, GitHub, VS Code
Learning now	Linux, computer networks, cloud computing
Planned	Docker, Terraform, CI/CD, Kubernetes

Licenses & Certifications (Planned)

- Foundation cloud certificate (such as AWS Cloud Practitioner or Azure Fundamentals) – target: before Year 2
- Associate-level cloud certificate – target: Year 2 or Year 3
- CKA or HashiCorp Terraform Associate – target: Year 3

Career Roadmap

Year	Focus	Deliverable
Year 1 (2026–27)	C, Python, Linux, networks, cloud basics	Static portfolio site hosted on the cloud
Year 2 (2027–28)	Docker, Python automation, Terraform	Containerised app deployed with Terraform
Year 3 (2028–29)	Kubernetes, CI/CD	App deployed through a CI/CD pipeline
Year 4 (2029–30)	Data structures, system design, capstone	Cloud capstone and placement preparation

Connect

GitHub: github.com/harikumarank | LinkedIn: linkedin.com/in/hari-kumaran-k-b74159428

Part B: GitHub profile README

Hi there, I’m Harikumaran K

CSE Student | Learning Linux & Cloud | Aspiring Cloud Engineer

LinkedIn: https://www.linkedin.com/in/hari-kumaran-k-b74159428
GitHub: https://github.com/harikumarank

About Me

- I’m a first-year Computer Science Engineering student building a base in programming, systems and the cloud.
- I’m working toward a cloud engineering career in a product-based company.
- I’m learning Linux and computer networks first, because cloud tools make more sense with those basics.
- I speak English, Tamil and Japanese.

Tech Stack & Tooling

Domain	Technologies & Tools
Languages	C, Python
Version control	Git, GitHub
Editor	VS Code
Learning now	Linux, computer networks, cloud computing
Planned	Docker, Terraform, CI/CD, Kubernetes

Featured Repositories

CloudGuard (planned): a Python tool that finds forgotten cloud resources and warns before a student’s budget is crossed.

Career Roadmap

- **Done:** started C and Python fundamentals.

- **In progress:** Linux, computer networks and cloud basics.
- **Goal:** a foundation cloud certificate before Year 2, then Docker, Terraform, CI/CD and Kubernetes.
- **Goal:** a cloud engineering role in a product-based company.

Certifications & Achievements

None yet. This section will grow as I complete certificates and projects.

Connect With Me

I’m happy to connect with fellow students and engineers interested in cloud and Linux. Reach out on LinkedIn: https://www.linkedin.com/in/hari-kumaran-k-b74159428